

Federated Learning-Based Diabetic Retinopathy Detection

SCAFFOLD + Personalization + Differential Privacy on IRDRD Dataset

Sarth Narola Nevil Anghan

Department of Computer Science and Engineering
Nirma University, Ahmedabad, India

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Sarth Narola · 23bce194@nirmauni.ac.in **Nevil Anghan** · 23bce195@nirmauni.ac.in

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Introduction: Diabetic Retinopathy (DR)

- DR is a serious diabetes complication that damages retinal vessels
- Progresses without symptoms — patients often notice late
- Leading cause of **preventable blindness** in working-age adults

Global Burden

- **103 million** people affected today
- Projected to reach **700 million by 2045**
- India is among the **top 3 nations** for diabetes burden

Five DR Severity Stages

- | | |
|----------------|------------------|
| Grade 0 | No DR |
| Grade 1 | Mild NPDR |
| Grade 2 | Moderate NPDR |
| Grade 3 | Severe NPDR |
| Grade 4 | Proliferative DR |

Key Challenge

Early stages are asymptomatic — automated screening at scale is urgently needed

Why Federated Learning for DR?

Problem with Centralized AI

- Deep learning needs large, diverse datasets
- Patient retinal images cannot be freely shared
- HIPAA (USA) and GDPR (EU) restrict cross-institutional data sharing
- Single central data store = single privacy risk

Federated Learning Solution

- Each hospital keeps patient data locally
- Only **model updates** (not images) are shared
- Global model is trained collaboratively
- Multi-hospital collaboration while staying compliant

Remaining Challenges in FL for DR

Standard FedAvg suffers from **client drift** on Non-IID hospital data $\Rightarrow$ slow, unstable convergence. Basic FL has no **formal privacy guarantees**. Most systems are not tested on the **Indian patient population**.

Problem Statement

Core Problem

Standard FL-based DR systems use FedAvg on assumed IID data, without formal privacy guarantees, and have not been tested on Indian patient populations using the IRDRD dataset.

Specific Issues:

- 1 FedAvg fails on **Non-IID** hospital data → client drift → poor convergence
- 2 No FL paper has applied **SCAFFOLD** to DR detection
- 3 Most FL systems have **no formal privacy** (DP) beyond gradient sharing
- 4 No federated study uses the **IRDRD dataset** (Indian population)
- 5 **Personalization** is rarely combined with privacy-preserving aggregation
- 6 No paper gives an empirical **FedAvg vs. SCAFFOLD** comparison on retinal data

Research Objectives

- 01.** Design a federated CNN framework for **5-class DR grading** on the **IRDRD dataset**
- 02.** Implement **SCAFFOLD** as the primary FL aggregation algorithm to handle Non-IID data
- 03.** Provide **Differential Privacy** (DP-SGD) for formal end-to-end privacy guarantees
- 04.** Add **Personalization** via local fine-tuning to adapt the global model to each hospital
- 05.** Conduct a direct **FedAvg vs. SCAFFOLD** comparison under identical Non-IID conditions
- 06.** Deliver a privacy-compliant system satisfying HIPAA/GDPR requirements

Literature Review: Overview (15 Papers, 2020–2025)

General Findings:

- FedAvg dominates: used in **12 of 15** papers
- Accuracy range: **79% – 98.6%**
- Datasets: IDRiD, EyePACS, APTOS, MESSIDOR, Kaggle DR
- Models: EfficientNet, ResNet, DenseNet, MobileNet, ViT
- Only **1 paper** (P12) uses Differential Privacy
- Only **1 paper** (P3) uses Personalization
- Only **1 paper** (P5) uses SCAFFOLD (not on medical data)
- **No paper** combines all three

Key Papers Summarized

Paper	Algorithm	Accuracy
P1 (2024)	FedAvg	93.21%
P3 (2024)	FedAvg+Personal.	94.66%
P5 (2020)	SCAFFOLD	Faster conv.
P7 (2022)	FedSGD	97.87%
P10 (2023)	FedAvg (mod.)	98.6%
P12 (2023)	FedAvg+DP	79.35%
P14 (2025)	FedProx	98.36%

Literature Review: All 15 Papers at a Glance

No.	Year	Model	FL Algorithm	Dataset	Accuracy / AUC	Key Contribution
P1	2024	EfficientNetB0	FedAvg	DDR, EyePACS, APTOS	93.21%	Multi-hospital FL; mobile deployment
P2	2025	ECSRNet (ShuffleNet+CSP+GRU)	FedAvg+IKMC+HE	Kaggle DR 2015	98.6%	Encryption + clustering in FL
P3	2024	Multiple CNNs	FedAvg + Personalization	IDRiD, STARE	94.66%	Client-level personalization in FL
P4	2023	Vision Transformer	FedAvg + ViT	APTOS, MESSIDOR-1/2, EyePACS	AUC: 0.979	First ViT in FL for DR
P5	2020	Control Variates	SCAFFOLD	Simulated, MNIST	Faster convergence	SCAFFOLD; solves client drift
P6	2022	AlexNet (TL)	FedAvg, FedProx	EyePACS, IDRiD, APTOS	FedAvg: 90.07%	Transfer learning + FL
P7	2022	CNN	FedSGD	Kaggle DR	97.87%	First FL-DR detection system
P8	2023	EfficientNet-B5	FedAvg (weighted)	OPHDIAT (~697k)	AUC: 0.9522	Real multi-center FL tested
P9	2025	CNN + GAN	FedAvg (weighted)	Kaggle, EyePACS, Messidor	~97%	FL + GAN augmentation; 5-class
P10	2023	Custom CNN	FedAvg + median CE loss	MESSIDOR, IDRiD, Kaggle	98.6%	DRFL; modified loss for FL
P11	2025	ResNet+ShuffleNet	FedAvg + XAI	APTOS, EyePACS	No results (proposal)	FL + TL + Grad-CAM + SHAP
P12	2023	CNN + DP-SGD	FedAvg + DP-SGD	Retina (35k, 5 classes)	79.35% (with DP)	Differential privacy in FL for DR
P13	2023	DenseNet	FedAvg + MPCE	APTOS 2019 (~5k)	Better than baselines	DenseNet in FL; dense feature reuse
P14	2025	MobileNet	FedProx + GFA	DDR (13,673 images)	98.36%	Lightweight FedProx; 20 clients
P15	2025	DNN (no FL)	Centralized only	Kaggle DR	Better than SVM	DNN vs. classical ML baseline

Research Gaps Identified

- ❶ **Gap 1:** SCAFFOLD has **never been applied to DR detection** or any medical imaging task despite proven Non-IID advantages (P5)
- ❷ **Gap 2:** No FL-based DR study uses the **IRDRD dataset** — Indian patient population is underrepresented
- ❸ **Gap 3:** Non-IID data heterogeneity is **largely ignored** — most papers assume IID or apply basic averaging
- ❹ **Gap 4:** **No single system combines SCAFFOLD + Personalization + Differential Privacy** — explored only in isolation (P3, P5, P12)
- ❺ **Gap 5:** No paper provides an empirical **FedAvg vs. SCAFFOLD comparison** on retinal fundus data
- ❻ **Gap 6:** Papers with DP (P12) lack SCAFFOLD; papers with personalization (P3) lack DP — **no end-to-end privacy-preserving personalized FL system for DR**

Gap Analysis: Feature Comparison Table

Feature	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14	P15	Ours
FedAvg	✓	✓	✓	✓	—	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	✓
FedProx	—	—	—	—	—	✓	—	—	—	—	—	—	—	✓	—	—
SCAFFOLD	—	—	—	—	✓	—	—	—	—	—	—	—	—	—	—	✓
Personalization	—	—	✓	—	—	—	—	—	—	—	—	—	—	—	—	✓
Differential Privacy	—	✓	—	—	—	—	—	—	—	—	—	✓	—	—	—	✓
Non-IID Handling	—	—	—	—	✓	—	—	—	—	—	—	—	—	✓	—	✓
Multi-center FL	—	—	—	—	—	—	—	✓	—	✓	—	—	—	—	—	✓
5-Class Grading	✓	✓	—	✓	—	—	—	—	✓	✓	—	✓	—	—	✓	✓
IRDRD Dataset	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	✓
Transfer Learning	—	—	—	—	—	✓	—	—	—	✓	—	—	—	—	—	—
GAN Augmentation	—	—	—	—	—	—	—	—	✓	—	—	—	—	—	—	—
XAI	—	—	—	—	—	—	—	—	—	—	✓	—	—	—	—	—

Highlighted rows = features that define our novelty contribution

Proposed System: Architecture Overview

Central Server

- Hosts global CNN model
- Maintains SCAFFOLD control variates c
- Aggregates Δw_k and Δc_k from clients
- Broadcasts updated w^{t+1} and c^{t+1}

Each Client (Hospital Node)

- Holds local IRDRD partition (Non-IID)
- Trains local CNN with DP-SGD
- Applies SCAFFOLD gradient correction
- Sends only weight updates (no images)

Three-Layer Privacy

- 1 **FL Level:** No raw images shared
- 2 **DP Level:** Gaussian noise on gradients ($\epsilon \leq 5$)
- 3 **Personalization:** DP applied during local fine-tuning too

System Stack

PyTorch + Flower (flwr)
Opacus (DP-SGD)
OpenCV + Albumentations
5 clients, 100 rounds
IRDRD dataset (Non-IID, $\alpha = 0.5$)

Step-by-Step Federated Workflow

- 1. Dataset Preparation** — IRDRD split into 5 Non-IID subsets via Dirichlet sampling ($\alpha = 0.5$)
- 2. Preprocessing (local)** — Resize 224×224 , CLAHE, normalize, augment
- 3. Model Init** — Server initializes CNN + SCAFFOLD control variates $c^0 = 0$
- 4. Local Training (DP-SGD)** — Gradient clipping + Gaussian noise; SCAFFOLD corrects gradient direction
- 5. Transmit Updates** — Send Δw_k and Δc_k only (no images)
- 6. Server Aggregation** — Aggregate weight and control variate updates
- 7. Broadcast** — Updated w^{t+1} , c^{t+1} sent to all clients
- 8.** Repeat for **100 rounds** or until convergence
- 9. Personalization** — Freeze feature extractor; fine-tune FC layers locally (10 epochs, DP-SGD)

CNN Model Architecture

Input: $224 \times 224 \times 3$ RGB retinal fundus image

Layer	Operation	Output Shape	Role in System
Block 1	Conv2D (32, 3×3)+BN+ReLU+MaxPool	$112 \times 112 \times 32$	Feature extraction
Block 2	Conv2D (64, 3×3)+BN+ReLU+MaxPool	$56 \times 56 \times 64$	Feature extraction
Block 3	Conv2D (128, 3×3)+BN+ReLU+MaxPool	$28 \times 28 \times 128$	Feature extraction
Block 4	Conv2D (256, 3×3)+BN+ReLU+MaxPool	$14 \times 14 \times 256$	Frozen during personalization
GAP	Global Average Pooling	256	Frozen during personalization
FC1	Dense (512, ReLU) + Dropout 0.5	512	Fine-tuned locally
FC2	Dense (256, ReLU) + Dropout 0.3	256	Fine-tuned locally
Output	Dense (5, Softmax)	5	Fine-tuned locally

Loss: Categorical Cross-Entropy Optimizer: Adam ($\eta = 1 \times 10^{-4}$)

SCAFFOLD Algorithm: How It Works

Problem: Client Drift in FedAvg

On Non-IID data, each client's local gradient pulls the model toward its own local optimum, not the global one. After E local epochs, models diverge significantly.

FedAvg Aggregation

$$w^{t+1} = \sum_{k=1}^N \frac{n_k}{n} w_k^{t,\text{local}}$$

Simple weighted average — does not correct drift.

SCAFFOLD: Corrected Gradient Step

Each local SGD step uses a corrected gradient:

$$w_k \leftarrow w_k - \eta_l (g_k - c_k + c)$$

c_k = local control variate, c = global control variate.

SCAFFOLD Server Update

$$w^{t+1} = w^t + \frac{\eta_s}{N} \sum_k \Delta w_k$$

$$c^{t+1} = c^t + \frac{1}{N} \sum_k \Delta c_k$$

Both weights and control variates are updated.

Differential Privacy and Personalization

Differential Privacy via DP-SGD

Applied during local training at each client:

- 1 Compute per-sample gradients $g_i(w)$
- 2 **Clip:** $\tilde{g}_i = g_i / \max(1, \|g_i\|_2 / C)$
- 3 **Add noise:** $\hat{g} = \frac{1}{B} (\sum_i \tilde{g}_i + \mathcal{N}(0, \sigma^2 C^2 \mathbf{I}))$
- 4 Update weights with noisy gradient

$$C = 1.0, \sigma = 1.1, \delta = 10^{-5}, \epsilon \leq 5$$

Personalization via Local Fine-Tuning

After global convergence:

- **Freeze:** Conv Blocks 1–4 + GAP
- **Fine-tune:** FC1, FC2, Output layers
- **Data:** Local IRDRD partition only
- **Duration:** 10 local epochs
- **LR:** Reduced to $\eta = 1 \times 10^{-5}$
- DP-SGD applied here too

Result: Global features + local calibration

Privacy guarantee: No raw images shared at any stage. DP-SGD ensures (ϵ, δ) -DP during both federated training and personalization.

Dataset: IRDRD — Indian Retinal Image Dataset for DR

Dataset Characteristics:

- High-resolution color fundus retinal photographs
- Annotated by certified Indian ophthalmologists
- Represents the **Indian patient population**
- Five-class ICDR severity labels (Grade 0–4)
- Includes class imbalance typical of clinical settings

Usage in Proposed FL System:

- Partitioned into $N = 5$ subsets (Dirichlet $\alpha = 0.5$)
- Each client: 70% train / 15% val / 15% test
- Weighted random sampling for class imbalance
- **Zero images transmitted to server**

Why IRDRD?

Gap addressed:	No existing FL-DR study uses IRDRD
Population:	Indian patients
Relevance:	India = top 3 diabetes burden
Generalization:	Better than Western datasets for India

Novel Contribution

First FL-based DR study using IRDRD as primary training dataset

FedAvg vs. SCAFFOLD: Controlled Comparison

Parameter	FedAvg	SCAFFOLD
Dataset	IRDRD Non-IID	IRDRD (same partitions)
Clients N	5	5
Dirichlet α	0.5	0.5 (identical)
Local epochs E	5	5
Communication rounds	100	100
CNN architecture	Identical	Identical
Differential Privacy	DP-SGD ($C = 1.0, \sigma = 1.1$)	Same DP-SGD
Control variates	Not used	c_k per client + c global
Aggregation	Weighted average of weights	Weighted avg. + Δc_k update
Metrics compared	Global accuracy, per-class F1, convergence curve	

Expected Results: FedAvg vs. SCAFFOLD

Expected Performance Range

Metric	FedAvg	SCAFFOLD
Global Accuracy	82–88%	88–94%
Convergence speed	Slower	Faster
Rounds to 85% acc	More	Fewer
Non-IID stability	Unstable	Stable

Effect of Differential Privacy

- Modest accuracy reduction expected: 3–6% vs. non-private baseline
- Consistent with P12 (79.35% with DP vs. 83.05% without)
- Formal guarantee: $(\epsilon \leq 5, \delta = 10^{-5})$ -DP

Effect of Personalization:

- Each client's local model expected to **outperform global model** on its own test set by 2–5%

Why SCAFFOLD is better here

Real hospital data is always Non-IID. FedAvg's client drift is the biggest obstacle to practical FL deployment — SCAFFOLD directly fixes this.

Advantages of the Proposed System

Over Centralized Learning

- No patient data leaves the hospital
- HIPAA and GDPR compliant by design
- Scales to any number of hospitals
- Reduces single-point data breach risk

Over Standard FedAvg

- SCAFFOLD corrects client drift
- Faster convergence on Non-IID data
- Provable convergence guarantees
- More stable across diverse hospitals

Over Prior FL-DR Systems

- First to combine SCAFFOLD + Personalization + DP
- First FL study on IRDRD dataset
- Full 5-class grading with privacy
- DP-SGD applied to both training and personalization

System Properties

- Open-source (PyTorch + Flower + Opacus)
- Reproducible experiments
- Modular: DP level tunable
- Real multi-hospital deployment

Conclusion

- DR is a major global health challenge requiring scalable, privacy-preserving automated detection
- Standard FL (FedAvg) is insufficient for real-world hospital settings due to Non-IID client drift
- The proposed system **combines SCAFFOLD, personalization, and differential privacy** in a single federated CNN framework for 5-class DR grading
- Evaluated on the **IRDRD dataset** — the first FL-based DR study to do so
- A controlled **FedAvg vs. SCAFFOLD comparison** will provide the first empirical evidence of SCAFFOLD's advantage in medical imaging
- Expected: **88–94% accuracy** with SCAFFOLD + personalization; formal ($\epsilon \leq 5, \delta = 10^{-5}$)-DP guarantee; privacy-compliant deployment

Future Work

Integrate Grad-CAM for model explainability; deploy in a real multi-hospital network; explore adaptive noise scheduling for a better accuracy-privacy tradeoff

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Thank You

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Questions?